

Solving Quadratic Equations

Quadratic Formula & Completing the Square · Algebra · Grade 9–11

Name: _____

Date: _____

Learning Objectives

- Apply the quadratic formula to solve quadratic equations with real and irrational solutions
- Rewrite quadratic equations in standard form before identifying coefficients a, b, and c
- Simplify radical expressions and split solutions into two distinct roots

Problems

1. Identify the values of a, b, and c for the quadratic equation, then apply the quadratic formula to solve it.

$$x^2 - 5x + 6 = 0$$

2. Use the quadratic formula to find the two solutions of the equation.

$$x^2 + 7x + 10 = 0$$

3. Solve the quadratic equation using the quadratic formula. Give your answers as fractions in simplest form.

$$2x^2 - 5x + 3 = 0$$

4. First, rewrite the equation in standard form (set equal to 0), then solve using the quadratic formula.

$$2x^2 - x = 6$$

5. Rewrite in standard form, then use the quadratic formula to solve.

$$3x^2 + 5x = 2$$

6. Use the quadratic formula to solve the equation. Simplify any radicals in your answer.

$$8x^2 - 4x - 1 = 0$$

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7. Solve the equation using the quadratic formula. Leave irrational answers in simplified radical form.

$$x^2 - 4x + 1 = 0$$

8. Rewrite the equation in standard form, identify a, b, and c, then solve using the quadratic formula. Simplify your answer fully.

$$8x^2 + 6x = -5$$

9. Solve the quadratic equation using the quadratic formula. Express any irrational roots in simplified radical form.

$$5x^2 - 3x - 1 = 0$$

10. Rewrite the equation in standard form, then solve using the quadratic formula. Simplify all radical expressions fully and reduce all fractions.

$$6x^2 = 11x + 10$$

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Solving Quadratic Equations — Answer Key

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Answer Key

1. Answer: $x = 3$ and $x = 2$

- Identify: $a = 1$, $b = -5$, $c = 6$
 - Apply the quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
 - $x = \frac{(5 \pm \sqrt{(25 - 24)})}{2} = \frac{(5 \pm \sqrt{1})}{2} = \frac{(5 \pm 1)}{2}$
 - $x = \frac{(5 + 1)}{2} = 3$ or $x = \frac{(5 - 1)}{2} = 2$
-

2. Answer: $x = -2$ and $x = -5$

- Identify: $a = 1$, $b = 7$, $c = 10$
 - Discriminant: $b^2 - 4ac = 49 - 40 = 9$
 - $x = \frac{(-7 \pm \sqrt{9})}{2} = \frac{(-7 \pm 3)}{2}$
 - $x = \frac{(-7 + 3)}{2} = -2$ or $x = \frac{(-7 - 3)}{2} = -5$
-

3. Answer: $x = \frac{3}{2}$ and $x = 1$

- Identify: $a = 2$, $b = -5$, $c = 3$
 - Discriminant: $(-5)^2 - 4(2)(3) = 25 - 24 = 1$
 - $x = \frac{(5 \pm \sqrt{1})}{4} = \frac{(5 \pm 1)}{4}$
 - $x = \frac{(5 + 1)}{4} = \frac{6}{4} = \frac{3}{2}$ or $x = \frac{(5 - 1)}{4} = \frac{4}{4} = 1$
-

4. Answer: $x = 2$ and $x = -\frac{3}{2}$

- Subtract 6 from both sides: $2x^2 - x - 6 = 0$
 - Identify: $a = 2$, $b = -1$, $c = -6$
 - Discriminant: $(-1)^2 - 4(2)(-6) = 1 + 48 = 49$
 - $x = \frac{(1 \pm \sqrt{49})}{4} = \frac{(1 \pm 7)}{4}$
 - $x = \frac{(1 + 7)}{4} = \frac{8}{4} = 2$ or $x = \frac{(1 - 7)}{4} = \frac{-6}{4} = -\frac{3}{2}$
-

5. Answer: $x = \frac{1}{3}$ and $x = -2$

- Subtract 2 from both sides: $3x^2 + 5x - 2 = 0$
 - Identify: $a = 3$, $b = 5$, $c = -2$
 - Discriminant: $25 - 4(3)(-2) = 25 + 24 = 49$
 - $x = \frac{(-5 \pm 7)}{6}$
 - $x = \frac{(-5 + 7)}{6} = \frac{2}{6} = \frac{1}{3}$ or $x = \frac{(-5 - 7)}{6} = \frac{-12}{6} = -2$
-

6. Answer: $x = \frac{(1 + \sqrt{3})}{4}$ and $x = \frac{(1 - \sqrt{3})}{4}$

- Identify: $a = 8$, $b = -4$, $c = -1$
 - Discriminant: $16 - 4(8)(-1) = 16 + 32 = 48$
 - Simplify $\sqrt{48} = \sqrt{(16 \times 3)} = 4\sqrt{3}$
 - $x = \frac{(4 \pm 4\sqrt{3})}{16}$
 - Divide each term by 4: $x = \frac{(1 \pm \sqrt{3})}{4}$
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**7. Answer: $x = 2 + \sqrt{3}$ and $x = 2 - \sqrt{3}$**

- Identify: $a = 1$, $b = -4$, $c = 1$
 - Discriminant: $16 - 4(1)(1) = 12$
 - Simplify $\sqrt{12} = \sqrt{4 \times 3} = 2\sqrt{3}$
 - $x = (4 \pm 2\sqrt{3}) / 2 = 2 \pm \sqrt{3}$
 - $x = 2 + \sqrt{3}$ or $x = 2 - \sqrt{3}$
-

8. Answer: No real solutions (discriminant is negative)

- Add 5 to both sides: $8x^2 + 6x + 5 = 0$
 - Identify: $a = 8$, $b = 6$, $c = 5$
 - Discriminant: $36 - 4(8)(5) = 36 - 160 = -124$
 - Since the discriminant is negative, there are no real solutions
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9. Answer: $x = (3 + \sqrt{29}) / 10$ and $x = (3 - \sqrt{29}) / 10$

- Identify: $a = 5$, $b = -3$, $c = -1$
 - Discriminant: $9 - 4(5)(-1) = 9 + 20 = 29$
 - $\sqrt{29}$ cannot be simplified further
 - $x = (3 \pm \sqrt{29}) / 10$
 - $x = (3 + \sqrt{29})/10$ or $x = (3 - \sqrt{29})/10$
-

10. Answer: $x = 5/2$ and $x = -2/3$

- Subtract $11x$ and 10 from both sides: $6x^2 - 11x - 10 = 0$
 - Identify: $a = 6$, $b = -11$, $c = -10$
 - Discriminant: $121 - 4(6)(-10) = 121 + 240 = 361$
 - $\sqrt{361} = 19$
 - $x = (11 \pm 19) / 12$
 - $x = (11 + 19)/12 = 30/12 = 5/2$ or $x = (11 - 19)/12 = -8/12 = -2/3$
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